7

Restorative materials and their relationship to tooth structure

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7.1 Introduction

Modern restorative materials can be classified in several ways, in terms of their retention (chemically adhesive, macro-, micro- or even nano-mechanical), their chemistry (e.g. resin-based vs. acid–base reaction, filler particles), or their clinical properties (e.g. aesthetics, strength, handling). It is essential that these materials are considered closely with the histological substrate to which they will adhere or with which they will interact, in order to understand the complexities of each system and their potential clinical uses.

This chapter will outline and discuss aspects of dental materials science to enable the reader to understand and appreciate the links with relevant histology and relate this to the clinical aspects of minimally invasive operative dentistry. Also discussed is dental amalgam, still a popular restorative material among many dentists worldwide, although clinical indications for its use are becoming more limited as treatment rationales change and adhesive materials improve. This text will require supplementation from suitable dental histology and detailed dental material science texts.

7.2 Dental resin composite

Dental resin composites are aesthetic, plastic adhesive restorative materials that consist of co-polymerized methacrylate-based resin chains embedding inert filler particles (conferring strength and wear resistance) and requiring a separate adhesive (bonding agent) to micro-/nano-mechanically bond them to either enamel or dentine, respectively. However, not all modern dental composites are based purely on this methacrylate resin chemistry (see Section 7.2.6). Therefore the term 'composite resin' is inappropriate and should not be used.

7.2.1 History

Resin composites have developed over the past 50 years, after the introduction of the acid-etch technique (Buonocore, 1955) and methacrylate monomers (Bowen's resin—Bis-GMA (1971); see Section 7.2.2).

7.2.2 Chemistry

Methacrylate resins

The unset (or uncured) material consists of a mixture of several different types of resin methacrylate monomers, most of which are hydrophobic (water-hating) in nature (see Figure 7.1).

The monomer chain length affects certain properties of the resin composite:

- *Viscosity (or flowability) of the material*. This is important in order to minimize voids trapped within the uncured composite during placement and packing within the depths of a cavity (the stiffer the consistency, the greater the risk of trapping air voids). The shorter the uncured monomer length (and therefore the lower the molecular weight), the lower is its viscosity. Often shorter-length, lower-molecular-weight methacrylate monomers form the basis of the resin chemistry of flowable resin composites, and other diluent molecules may be added.
- *Volumetric polymerization shrinkage of composite on curing*. The setting process is a light-activated, free-radical addition polymerization chain reaction. The shorter the monomer chains, the more of them need to join per unit volume, and the increased reduction in space between them leads to the relatively increased shrinkage exhibited (1.5–3.5% by volume; see Figure 7.2).

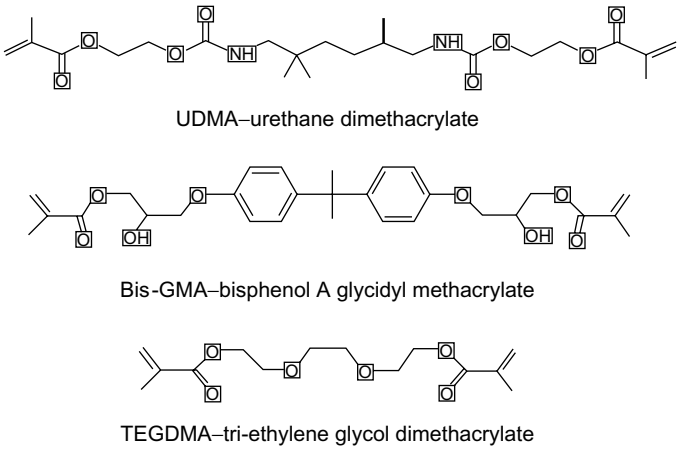


Figure 7.1 Examples with chemical formulae of methacrylate resins commonly used in dental composites. Note how TEGDMA has a shorter chain length than the others.

Q7.1: What properties of the resin composite are affected by monomer chain length?

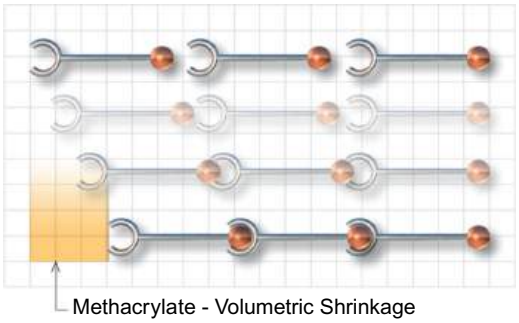


Figure 7.2 A diagrammatic representation of linear monomers undergoing addition polymerization leading to volumetric shrinkage. (Courtesy of 3M ESPE.)

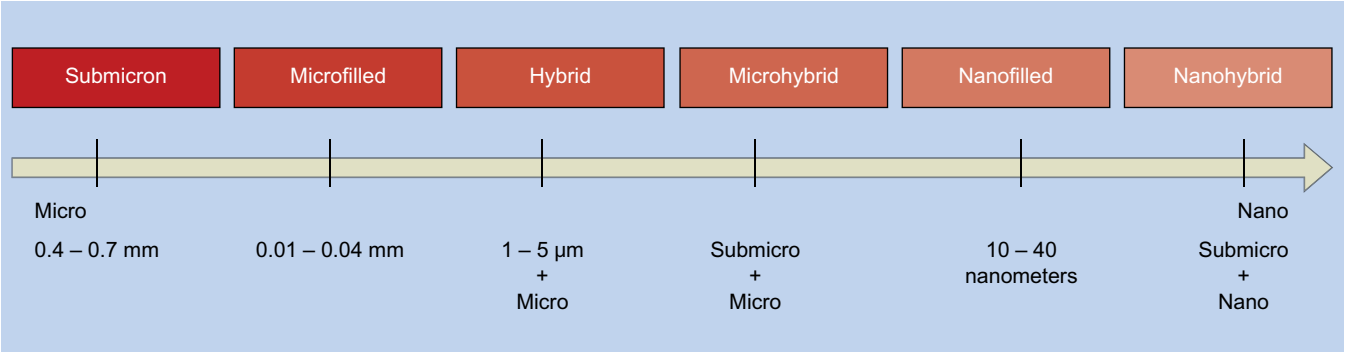


Figure 7.3 A table showing the size ranges of filler particles in a range of modern resin composites. (Courtesy of Septodont UK.)

Filler particles

See Figure 7.3. Inert filler particles are made from silica, quartz, barium, or strontium glass derivatives (introducing radio-opacity), and are embedded in and bound to the resin matrix using a coating of an organo-silane coupling agent (γ-methacryloxypropyltrimethoxysilane (γ-MPTS)). Over the past 40 years, manufacturers have attempted to pack in various sizes/shapes of particle in increasing numbers per unit volume in an effort to improve the clinical properties of resin composites. The more space that is taken up by irregular and spherical-shaped filler particles, the less space there is for native resin monomer, leading to a reduction in overall shrinkage on curing. Conversely, the more filler particles that are loaded into the composite, the more viscous it becomes, so there is a fine balance to be struck between the filler load and its resin content. The filler particles confer several properties on the cured composite material:

- *Wear characteristics*: increasing the density of larger, harder, and more irregular shaped filler particles will tend to increase the wear resistance of the composite, as less resin matrix will be exposed at the restoration surface.
- *Surface polish*: the smaller, softer, and more spherical-shaped the filler particles, the more polishable (but also less wear resistant) the resin composite.
- *Aesthetics*: the finer the filler particles, the more aesthetic the composite, as the optical properties can be made to match more accurately those of enamel. Modern resin composites can have numerous shades (using pigments and various degrees of opacity) or just a few, relying on a 'chameleon' effect where a relatively translucent resin composite can encompass several tooth shades, blending into the surrounding natural colours very effectively by transmitting colour from them.
- *Physical properties (e.g. compressive and shear strength, elasticity)*: the correct balance between filler particle and resin content is required to ensure optimal physical properties.

Other chemical ingredients

These small fractions include *inhibitors* (butyl-4-hydroxytoluenes (BHT), to increase material longevity prior to curing), *photo-initiators* (see later), *accelerators* (dimethylaminobenzoates (DMAB), to increase the reactivity of the photo-initiator, thereby speeding up curing time), *photo-stabilizers* (2-hydroxy-4-methoxybenzophenone (HMBP), to provide colour stability by eliminating UV action on amine initiators), *radio-opacifiers* (aluminium, titanium, or zirconium oxides), and *colour pigments* (various ferric and titanium oxides).

Setting reaction

The setting (curing) process in modern resin composites is either light-activated (470 nm wavelength visible blue light) or dual-cured, light and chemical activation (initiated by 0.5% benzoyl peroxide and activated using dimethyl-p-toluidine). Light-cured resin composite contains a photo-initiator, α-diketone (camphorquinone) and an amine, which under the activation of visible blue light generates the free radicals required to initiate the cross-linking polymerization chain reaction. Some issues related to the setting reaction of resins include the following:

- *Low degree of monomer conversion (50–77%)*: not all of the free monomer polymerizes on curing, leaving some to leach out of the composite, so contributing to the long-term degradation of the material in terms of moisture ingress and possible sensitization issues with some patients.
- *High linear/volumetric shrinkage on curing (1.5–3.5%)*: Figure 7.2 illustrates how shrinkage occurs during polymerization and leads to the composite pulling away from cavity margins, so increasing the risk of marginal leakage.
- *High shrinkage stress at the composite–tooth interface (3–8 MPa)*: the stresses generated by the polymerization process can be high at cavity margins, depending on the cavity C factor (the ratio of bonded to unbonded surfaces; see Chapter 6), the bulk of the material, and the compliance of the cavity walls. This can lead to debonding and tooth fracture of the weakened marginal tissues.
- *Water absorption*: water is absorbed hygroscopically during and after the curing process, leading to expansion, long-term composite degradation, and staining.
- *Air-inhibited surface layer of uncured resin*: the free-radical addition polymerization reaction is inhibited by air. After curing an increment of resin composite material, a glossy film of uncured resin is retained

on the surface of the composite, and the monomers in this layer are used to provide adherence for the next increment placed upon it. In this way, resin composite can be added to within a cavity to build up the final restoration.

- *Depth of cure*: conventional resin composites can be cured to a depth of 2–3 mm (depending on the translucency of the material). A thicker section will not cure sufficiently at its base (the 470 nm light does not penetrate sufficiently at those depths to permit activation of the polymerization reaction), and the resulting restoration may then be described as having a 'soggy bottom.' Note that resin composites cure most efficiently when closest to the 470 nm light source, which must be placed as close as possible to the uncured composite and should be built up in small angled increments.
- *'Bulk fill' resin composites:* these materials are promoted by certain manufacturers as being suitable for use as a dentine replacement in the restoration of large deep cavities. They are usually of a relatively low-viscosity, 'flowable' consistency so that they will achieve good adaptation to the cavity floor and walls, almost 'self-levelling' within the cavity. Mechanical properties, such as toughness, will be reduced to achieve this effect, but this may be an acceptable compromise within the depths of a large cavity. Their depth of cure is achieved by the use of a relatively transparent material, often with large filler particles, so allowing good light transmission and hence monomer conversion. There are also more viscous, heavily filled versions of these materials, which can be placed in bulk, but there are compromises in their handling, such as achieving adequate adaptation on the cavity floor (leading to voids), good contact points, and the development of occlusal anatomy.

7.2.3 The tooth–resin composite interface

Table 7.1 highlights some of the clinically important histological features of sound enamel and dentine relevant to the restoration of tooth surfaces with modern adhesive materials. An understanding of the interactions between the chemistry of the materials and this relevant histology is vital in order to optimize the qualities of the adhesive restorative systems available, including resin composites. Table 7.2 outlines the clinical properties that each layer of dental hard tissue confers on the crown of a tooth, and those properties that a restorative material should exhibit to act as an ideal replacement material for each.

Resin composite and enamel

Hydrophobic resin composites adhere to dry enamel micro-mechanically, and the prismatic, crystalline ultrastructure of enamel permits this to happen successfully, reliably, and strongly (20–50 MPa bond strengths have been measured in laboratory studies). Three stages are required to develop the bond between resin composite and enamel after the enamel has been prepared mechanically during cavity preparation:

1. Acid etching (alternative US term is 'conditioning')—this is the process of placing 37% orthophosphoric acid gel on to the prepared enamel surface for 20 seconds. This:
   - removes surface contaminants (saliva, proteinaceous substances) and the smear layer (a tenacious surface layer of organic and inorganic cutting debris < 20 µm thick) and increases the surface area for bonding
   - produces micro-irregularities/micro-porosities in the prismatic enamel surface. This microscopic surface roughness will provide micro-mechanical retention for the resin (see Figure 7.4).

Table 7.1 Clinically relevant histology that interacts with the chemistry of modern adhesive restorative materials

	Mineral (inorganic)	Matrix (organic)	Structural arrangement
Enamel	Calcium hydroxyapatite crystallites ($Ca_{10}(PO_4)_7(OH)_2$: 40 × 70 × 170 nm) 97% by weight, 89% by volume	No collagen Enamelins (MW 55 kDa), amelogenins (MW 25 kDa): 1% by weight, 2% by volume Water: 3% by weight, 9% by volume	10 000 crystallites arranged into *prisms* (US-rods); EDJ to surface; ameloblasts—Tomes' processes; 4–7 µm diameter; undulation, decussation; keyhole pattern in cross-section; prism cores/boundaries exist due to change in orientation of crystallites Aprismatic layer at tooth surface on eruption
Dentine	Calcium hydroxyapatite crystallites (with Mg and CO_3 substitutions: 5 × 30 × 80 nm) β-octocalcium phosphate crystals	90% collagen: Type I ($[\alpha1(I)]_2\alpha2(I)$) with trace amounts of type I trimer ($[\alpha1(I)]_3$), 300 nm rod-shaped triple helical collagen molecules linearly aligned and cylindrically grouped as fibrils. Parallel fibrils gathered into fibres 10% non-collagenous proteins (phosphorylated phosphoproteins (MW 140 kDa), Gla-proteins, macromolecular proteoglycans, plasma proteins, acidic glycoproteins), water	*Tubules* (EDJ to pulp chamber); approximately 19–45,000 per mm^2; coronal sigmoid curvature; 1–5 µm diameter; anastomosing branches evident (0.5 µm to 25 nm diameter); odontoblasts and processes

MW, molecular weight; kDa – kilodaltons; EDJ, enamel–dentine junction.

Table 7.2 Clinical properties that each of the dental hard tissues confers on a tooth, and those properties that an ideal restorative material should mimic

	Enamel	Enamel–dentine junction	Dentine
Clinical properties	Rigidity/brittle High compressive strength Wear resistance Translucency Dry	Scalloped Optical tint Limits crack propagation/ catastrophic failure Approximately 50 MPa bond strength	Bulk Slight elasticity/flexibility Shade Dynamic hydration: wetter, closer to the pulp (affects bonding ability)
Restorative replacement material	Dental resin composite	Dentine bonding agent Dynamic chemical bond	Glass ionomer cement

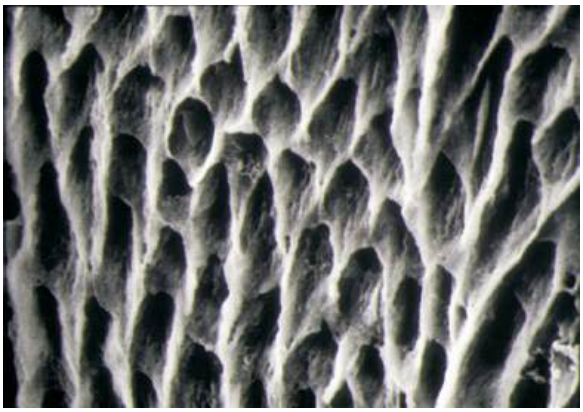


Figure 7.4 The tooth's view of an acid etch-retained resin bond. The enamel was etched, washed, bonded, and then dissolved away using a strong acid. The acid-resistant resin layer was retained. It can be seen to have flowed into the prism boundary regions (field width 80 µm). (Courtesy of Professor A Boyde.)

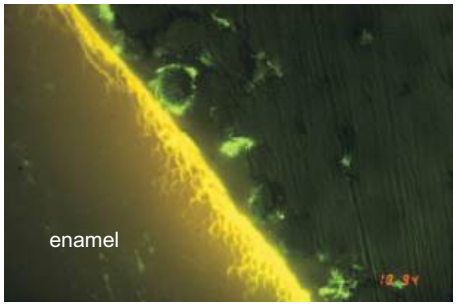


Figure 7.5 The interface between etched and bonded enamel prisms and a restoration. The bonding agent has been labelled with a yellow fluorescent dye (fluorescein), and its penetration around the horseshoe-shaped enamel prisms can be seen. Enamel prisms are easily separated from one another in the lateral walls of a cavity such as this, especially if the restoration shrinks (field width 500 µm).

2. After washing the acid etch gel off and drying the enamel thoroughly, a layer of fluid unfilled hydrophobic bonding resin (i.e. the resin without filler particles) is placed on to the etched enamel surface. This resin can flow easily into the micro-undercuts created by the etching process, and it is then light cured. Note that the chemistry of the unfilled fluid bonding resin is not the same as that of a dentine bonding agent (see later).
3. The final composite can then be placed on the layer of photo-cured bonding resin, in small angled increments (to lessen the effects of polymerization shrinkage), and photo-cured to form the final contoured restoration (see Figure 7.5).

When bonding to enamel, consideration must be given to the quality of the prismatic structure, as this is critical to adhesion. Unprepared young enamel may still retain a layer of *aprismatic* enamel—that is, the final layers of enamel to be laid down before completion of enamel development. This layer is void of enamel prisms as the elongated Tomes' processes of the enamel-producing cells, the ameloblasts, had already retracted as enamel deposition was nearing completion. Up to 10 microns in thickness, this layer is often removed in normal function due to the normal causes of tooth surface loss (erosion, abrasion, and attrition; see Chapter 1, Section 1.3). Prisms that have been sectioned/cracked/pulled apart during tooth preparation with rotary instrumentation, and whose bases do not reach the enamel–dentine junction (EDJ) intact, are described as being *unsupported*. These prisms are inherently weak, and if bonded to they will pull apart physically under the stresses exerted on them by the curing resin composite. This will lead to relatively short-term marginal failure within the enamel (i.e. cohesive enamel failure). In theory, if supported prisms are cut at, or close to, 90° to their long axis (i.e. clinically, the enamel margin is lightly bevelled) then the deleterious effects of shrinkage stress at the margins may be reduced, as prisms possess greater tensile

and compressive strength along their c-axes, as opposed to perpendicular to the c-axes. Enamel that has lost much of the underlying dentine beneath it (*undermined* enamel) will need to have this replaced with a suitable material prior to the resin composite–enamel bond being created.

Resin composite and dentine: dentine bonding agents

The fundamental clinical problem is to achieve intimate compatibility of two immiscible substrates—dentine is hydrophilic (likes water and is innately wet), whereas methacrylate resins are hydrophobic (water-hating). Therefore a *dentine bonding agent (DBA)* is required as an adhesive between the two. DBAs have three generic components:

- *Etch*: the first stage of the process, which has the following effects on dentine:
  - It removes the dentine smear layer created by the cavity preparation process (see Figure 7.6).
  - It helps to unblock and widen the dentine tubule orifices.
  - It demineralizes the dentine surface, thereby exposing the network of collagen in the dentine matrix (see Table 7.1).
- *Primer*: a bifunctional coupling molecule (e.g. hydrophilic HEMA—hydroxyethylmethacrylate) which has one functional group that likes water, so is compatible with the moist collagen in dentine, and another that likes resin, so is compatible with the bond/composite. The primer is carried into the moist collagen fibrillar network using a solvent (water, acetone, or alcohol), displacing the water molecules from the collagen and permitting the primer to enter the micro- and nano-spaces created around the collagen fibrils after the etching process. Other monomers that have been shown to promote successful bonding to hydroxyapatite and calcium ions include the MDP monomer (10-methacryloyloxydecyl dihydrogen phosphate). This has a terminal double bond group for polymerization, a hydrophobic alkylene group to maintain the balance between hydrophobic and hydrophilic properties, and a hydrophilic phosphate group for acid demineralization and chemical bonding to tooth structure.
- *Bond*: this can be considered simply as the resin component of the composite without the filler particles. A mixture of low-viscosity monomers able to penetrate the spaces not occupied by the primer monomers, it can be cured in the conventional way and then finally the resin composite placed incrementally (chemically cross-linking to the monomers in the air-inhibited layer of the bond) and the restoration completed.



Figure 7.6 Dentine smear layer (a tenacious layer of organic and inorganic cutting debris usually < 20 µm thick) imaged using scanning electron microscopy: this can be a barrier to good bonding, and is normally removed or modified using acids (field width 500 µm).

Q7.2: A dentine smear layer is illustrated in Figure 7.6. How is the smear layer created?

Each of these sensitive stages needs to be accomplished in optimal clinical conditions to enable a successful bond to be created between resin composite and dentine. The status of the collagen network on the cut dentine surface is of prime importance—it is the penetration of the primer and bond into these micro-/nano-spaces that provides the integrity of the seal and bond between resin composite and dentine. This zone of collagen penetration is known as the *hybrid zone*, and it can be 0.5–15 µm thick. Note that the thickness of the hybrid zone is less important than its continuity and integrity along the bonding interface. Any significant gaps in the hybrid zone will reduce the quality of the seal and ultimately of the bond. Penetration of the bonding agent into the dentine tubules will also play a part in reinforcing the bond strength (see Figure 7.7). DBAs can also be used on enamel instead of the unfilled bonding resin described earlier, to simplify the overall bonding process, but note that the chemistries of unfilled bonding resin (for enamel only) and DBAs (for both dentine and enamel) are different. There are numerous clinical presentations of DBAs from different manufacturers, and a simple classification system dependent on the three stages described previously is presented in Table 7.3. In Chapter 8, Table 8.12 outlines in more detail the differences in the clinical techniques employed for each type of system.

7.2.4 Classification of dentine bonding agents

Type 1 (three-bottle or three-step bonding systems)

Each of the three clinical steps is completed separately. Ideally, dentine should be etched for 10–15 seconds and enamel for 20 seconds using 37% orthophosphoric acid, which is then washed away thoroughly (for 10 seconds) and dried until the enamel appears frosty white. The primer contains water, so the over-dried exposed collagen on the dentine surface will rehydrate up to a point, thus permitting the primer and bond to penetrate the collagen network. However, over-etching coupled with incomplete rehydration of the collagen might leave unfilled

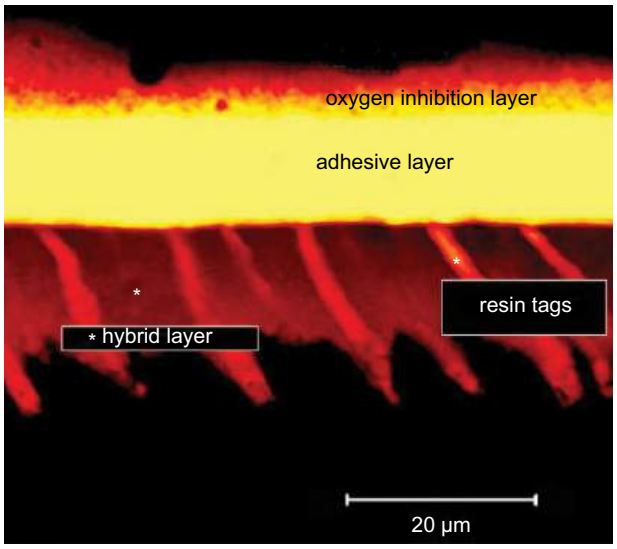


Figure 7.7 A fluorescent micrograph of a dentine bonding agent, showing penetration of the red-labelled (rhodamine) primer into the collagen network and tubules in dentine, forming the hybrid zone up to 15 µm thick (nano-retention) as well as resin tags (micro-retention). The yellow-labelled (fluorescein) adhesive uses its oxygen-inhibition layer to aid adherence to the composite. (Courtesy of S. Sauro.)

micro-/nano-porosities within the depth of the hybrid zone, and subsequent fluid movement might lead to post-operative sensitivity and bond degradation over time. Even though the clinical technique sensitivity is greater for these type 1 systems, due to the three separate steps, it is universally accepted that type 1, three-bottle systems provide a good-quality and reliable bond, and they have been used for many years as the gold standard for dentine bond measurement in laboratory studies.

Type 2 (two-bottle or two-stage, 'total etch' or 'etch and rinse' adhesives)

The clinical stages have been simplified into two steps—an initial etch, which is then rinsed away thereby removing the smear layer. In type 2 systems it is imperative not to over-dry the exposed collagen network, as this cannot be rehydrated by the contents of the second bottle containing both the primer and the bond (see Figure 7.8). A technique of 'moist bonding' must be employed after washing off the etchant gel, wicking the 'puddles' of excess water away using cotton wool pledgets, paper points, or very gentle air drying (see Chapter 8). This will ensure that the collagen network remains erect and supported, so more conducive to bond penetration (see Figure 7.9). The primer and bond are combined, and it is essential to evaporate the solvent carrier (acetone or alcohol based), as any remnants will contaminate and weaken the final bond. This procedure also thins the adhesive on the cavity walls, and the operator must ensure that there is enough adhesive present before the photo-curing stage (the cavity walls must appear shiny). Therefore multiple applications of the adhesive (primer and bond together) may be required prior to placing the resin composite restoration.

Type 3 (weaker self-etching primers)

The acid phosphate monomers in the primer act as the etchant, so no separate etching/rinsing/drying phase or moist bonding technique is required. The dentine is etched sufficiently to partially dissolve the smear layer and expose the collagen network. However, uncut

Table 7.3 A classification system for modern dentine bonding agents indicating the effect on the smear layer, and presentation options of the three stages—etch, primer, and bond—with some UK market examples

Classification of DBAs					
Type	**Smear**	**Etch**	**Prime**	**Bond**	***Examples***
1 three bottle (4th generation)	Remove				Adper Scotchbond MP Optibond FL
2 etch & rinse (5th generation)	Remove				Optibond Solo Prime & Bond NT/XP One Step Plus Scothbond Universal
3 mild self-etch (6th generation)	Dissolve				Clearfil SE Bond/Protect Bond
4 strong self-etch (7th generation)	Dissolve				Scotchbond Universal Xeno III; I Bond One up bond F Plus
		Milder self-etching primer			Fuji Bond LC; G Bond; Tri S Bond

A box that extends across two columns represents different components of dental adhesives being blended together.

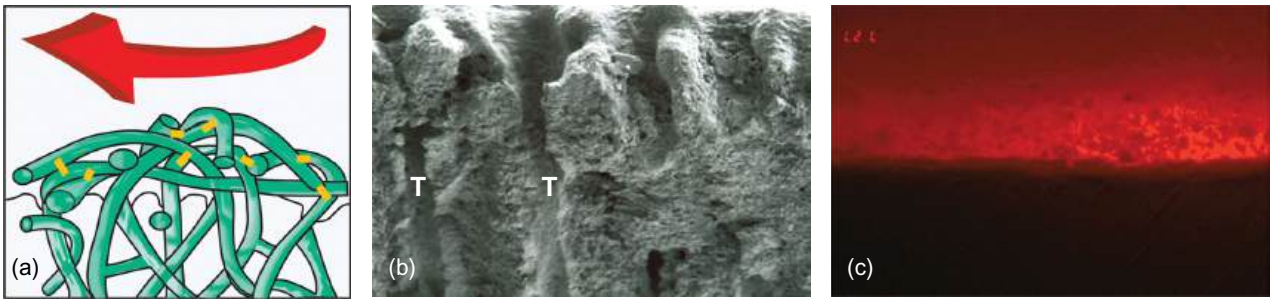


Figure 7.8 **(a)** Diagram showing collagen fibres in dentine that have been over-dried after etching. They have collapsed, so neither the primer nor bond can penetrate into the nano-spaces within the collagen network (red arrow). **(b)** A scanning electron micrograph (field width 75 µm) showing an over-dried cut dentine surface with a collapsed collagen network (T, dentine tubule cut longitudinally). **(c)** Fluorescent confocal microscopic image (field width 250 µm) showing the bonding agent labelled with rhodamine (red upper portion). The resin composite can be seen in the top half of the image and the very faint outlines of dentinal tubules in the lower half. Note that as the dentine has been over-dried (see Figure 7.8a), there has been no penetration of the collagen network or tubules, and the resulting bond and seal will be poor. (Figure 7.8 (a) courtesy of D. Ziskind.)

Q7.3: What are the dark irregular shapes outlined within the rhodamine-labelled (red) upper portion in Figure 7.8 (c)?

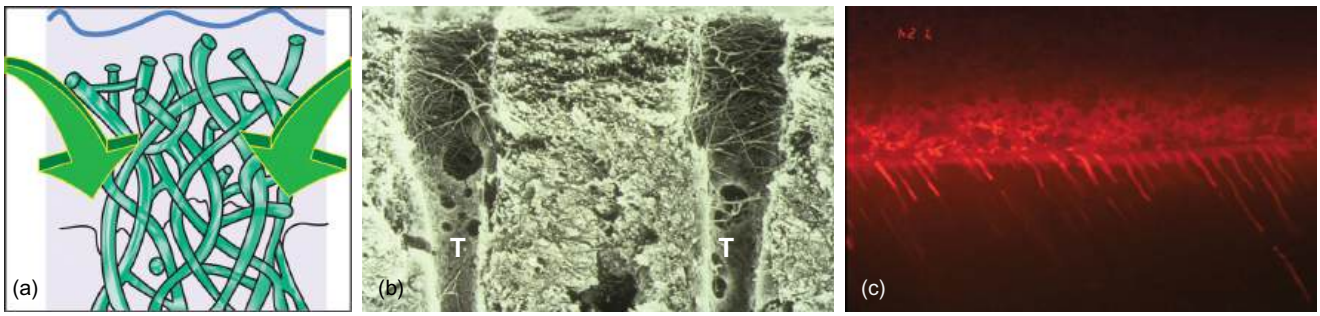


Figure 7.9 **(a)** Diagram showing collagen fibres in dentine that have been kept moist, so the network is upright with micro-/nano-spaces available to accept the primer and bond (green arrows). **(b)** An environmental scanning electron micrograph (field width 75 µm) showing moist collagen on the cut dentine surface. Individual collagen fibrils can be easily detected, and the orifices of two tubules have been clearly widened by the etching process (T, dentine tubule cut longitudinally). **(c)** Fluorescent confocal microscopic image (field width 250 µm) showing the bonding agent labelled with a red dye penetrating the moist collagen network on the dentine surface (nano-mechanical retention) and into the tubules (micro-mechanical retention). Note the thin red diffuse band at the interface between dentine and composite (arrow); this is the *hybrid zone* (10 µm thick). (Figure 7.9 (a) courtesy of D. Ziskind. Figure 7.9 (b) courtesy of Bart van Meerbeek.)

enamel margins may be etched less efficiently, leading to a distinct possibility of stained margins of restorations in the long term (see Figure 7.10).

This problem may be prevented by pre-etching the enamel margins with 37% orthophosphoric acid for 20 seconds, converting the whole clinical process to one similar to a type 1, three-stage system (see Chapter 8). The solvent again must be evaporated after the acidic primer is rubbed on to the dentine surface as in the type 2 systems.

Type 4 (stronger self-etching all-in-one systems)

Clinically, these are the simplest to use as there is a single application step, but the three bonding stages (etch, prime, and bond) are somewhat compromised. These are the most recent additions to the market, and laboratory studies on sound enamel and dentine show not only favourable bond strengths but also the presence of water blisters in the adhesive layer as it acts as a semi-permeable membrane, drawing water from the wet dentine tubules. As hydrolysis of polymeric dentine bonds occurs, there is evidence that marginal discoloration is more of an

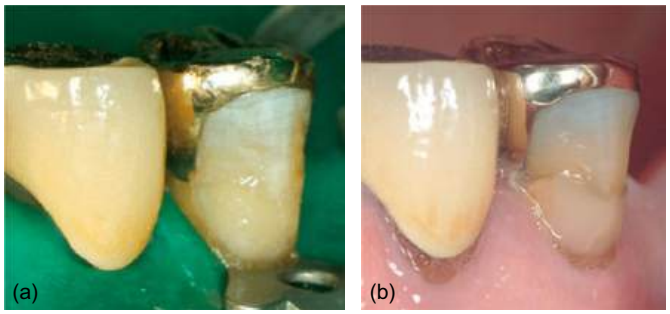


Figure 7.10 **(a)** Resin composite class V buccal cervical restoration LL5 placed using a type 3 self-etching primer dentine bonding agent. **(b)** The same restoration, 10 years later.

Q7.4: Look at Figure 7.10 (a) and (b). Can you spot the difference between the two and explain the reason for the difference?

issue with these adhesives. However, vigorous evaporation of the water-based solvent in the adhesive may reduce this problem.

7.2.5 Clinical issues with dentine bonding agents

- *Technique sensitivity*: The steps involved in dentine bonding require high levels of moisture control intra-orally, and appropriate handling of the dental materials themselves. For example, if volatile solvents are allowed to evaporate naturally from bottles of adhesive when the lids are not replaced after dispensing, the chemistry is irreversibly adversely affected, so bonding will be impaired. Primers and bond need to be agitated well into the tooth surface, and all these steps must be followed to obtain a successful bond. Avoid the temptation to take clinical short cuts and omit stages of the bonding procedure!
- *Packaging instructions*: Familiarization with the steps in the bonding process is essential, and teamwork with the nurse is vital in order to place the various materials in the correct order, with speed and precision. Single-dose capsules/foils/bubble-packs can be of assistance, reducing wastage and improving infection control procedures.
- *Shelf-life*: If kept refrigerated, the shelf-life of most DBAs is good. Volatile solvents need to be kept tightly stoppered, and as the materials are light sensitive, the use of blacked-out bottles is advised.
- *Suitable substrate*: Most studies on DBAs use sound enamel and dentine as the tooth surface to which to bond, often on extracted teeth. The histological properties of teeth *in vivo* may be quite different, and also the quality of the hard tissues will vary depending on the amount of carious tooth structure that has been retained. Therefore a comprehensive understanding is required of the state of the collagen network and mineral prior to using the DBA. More recent publications have used caries-affected dentine substrates *in vitro* and show clinically acceptable bond strengths (15–28 MPa).
- *Hydrolysis*: This is the major factor in the long-term degradation of the bond interface. The residual monomers will be affected by the water transition through vital dentine, thus ultimately compromising the seal and bond. Therefore a good-quality peripheral enamel seal is essential for the long-term success of adhesive bonding. Matrix metalloproteinases (MMPs) released from the exposed collagen (see Chapter 1), and activated in the caries acidic environment, may account for the failure of the adhesive bond in both caries-affected and sound dentine by accelerating collagen fibril cleavage. This in turn will cause a breakdown of the hybrid zone, thus jeopardizing the long-term adhesive seal and bond.
- *Sensitization*: These chemicals are designed to penetrate living tissue, and latex or nitrile gloves do not offer protection from resin-based materials. Therefore the dentist and nurse need to follow a 'no-touch' regimen when handling these materials.

7.2.6 Developments

The world of dental materials is constantly changing, and by the time this book is published new materials will be available for use clinically. Some of the challenges and developments are discussed in this section.

Reducing shrinkage stress and strain

Manufacturers can alleviate the problem of significant volumetric shrinkage (often leading to increased marginal stress and/or material strain) by producing low-shrink materials, or materials that impart little strain on the tooth. Such a system is illustrated by siloxane-oxirane (silorane) chemistry, as opposed to conventional methacrylates. In this system, the monomers, instead of being linear, are ring-opening and therefore, during the cationic polymerization process, less shrinkage occurs (0.9%). This chemistry requires a separate bonding agent, which is very hydrophobic, thereby minimizing the water transition through the adhesive. Another system is a dimer-acid nano-hybrid composite (based on dimer dicarbamate dimethacrylates), which exhibits polymerization-induced phase separation (expansion) on curing, thereby reducing some of the effects of polymerization shrinkage. Although these low-shrink materials have been shown to be effective, dentists are still using conventional resin composites, perhaps indicating that the problems of shrinkage stress can be mitigated by careful handling. Conversely, the 'bulk-fill' material developments (see Section 7.2.2) indicate that low stress/low viscosity composites that simplify restoration placement can become popular—time will tell if these are successful clinically.

Dentine bonding agents

With the advent of low-shrink composites, the importance of direct bond strengths may lessen and more emphasis may be placed on the ionic interaction between the materials and tooth structure. Medicinal ions could be transferred from the adhesive to help to remineralize the enamel or dentine, or act as anti-bacterial agents.

'Self-adhesive composite'

The ultimate development, a simple to use, self-adhesive composite restorative material, has been marketed, but bond strengths to dental hard tissues are not yet good enough for it to be recommended for regular use. The technology will not be long in the offing, however, as clinically successful self-etch resin cements do exist, currently used to adhere indirect restorations and root canal posts to teeth.

7.3 Glass ionomer cement

Glass ionomer cement (GIC) is a water-based, plastic direct dental restorative cement formed from an acid–base reaction between a polyalkenoic acid and ion-leachable fluoro-calcium (strontium) aluminosilicate glass particles.

7.3.1 History

GIC was developed as a biocompatible, chemically adhesive plastic restorative material by Wilson and Kent in the UK in 1972.

7.3.2 Chemistry

- *Powder*: calcium fluoro-aluminosilicate glass particles with strontium (to increase radio-opacity). The silica (SiO_2—affects transparency), alumina (Al_2O_3—affects opacity and setting time, and can increase the compressive strength of the set cement), and calcium fluoride (CaF_2—the fluoride ions reduce the fusion temperature, increase the strength of the set cement, enhance translucency, and have a therapeutic effect) are the important components. These are structured ionically as a tetrahedral complex with a centrally located aluminium ion and closely localized alkaline earth cations (sodium, potassium, calcium, and strontium) to maintain electro-neutrality.
- *Liquid-polyacid*: itaconic acid copolymer solution in water plus tartaric acid (5–15%, maintains working time and assists setting reaction). Anhydrous forms have vacuum-dried polyacrylic acid incorporated into the powder and are mixed with water ± dilute solution of tartaric acid.
- *Setting reaction*: acid–base reaction with three stages:

1. *Dissolution*: sol formation from the outer layers of the glass particles as they are attacked by the polyacid, and calcium, strontium, aluminium, and fluoride ions are released.
2. *Gelation and hardening*: primarily calcium ions bind to carboxylate groups (4 to 5 minutes) producing a clinically hard surface (initial set), and a silaceous hydrogel is formed, maturing over 24 hours, with further cross-linking with aluminium ions, up to 7 days. This process causes volumetric shrinkage of up to 3%. Surface protection at this stage is advisable to ensure regulation of the movement of water molecules in and out of the hydrogel, so permitting maturation to occur.
3. *Hydration*: associated with Stage 2, this continues over 4 to 6 months, so improving the physical properties of the material. The uptake of water leads to expansion, thereby reducing the ill effects of the initial shrinkage.

7.3.3 The tooth–GIC interface

GICs have the ability to adhere chemically to mineralized dental tissues. They do this by the dynamic processes of diffusion and adsorption.

Enamel

GIC polyacid displaces phosphate and divalent Ca^{2+} ions from the hydroxyapatite, and these ions become incorporated into the GIC matrix, which sets. Subsequently the pH rises and re-precipitation of minerals at the GIC–tooth interface occurs, forming an ion-enriched layer, which is so firmly bound that when GIC restorations fail, this can be cohesive within the GIC material itself (see Figure 7.11). If the enamel

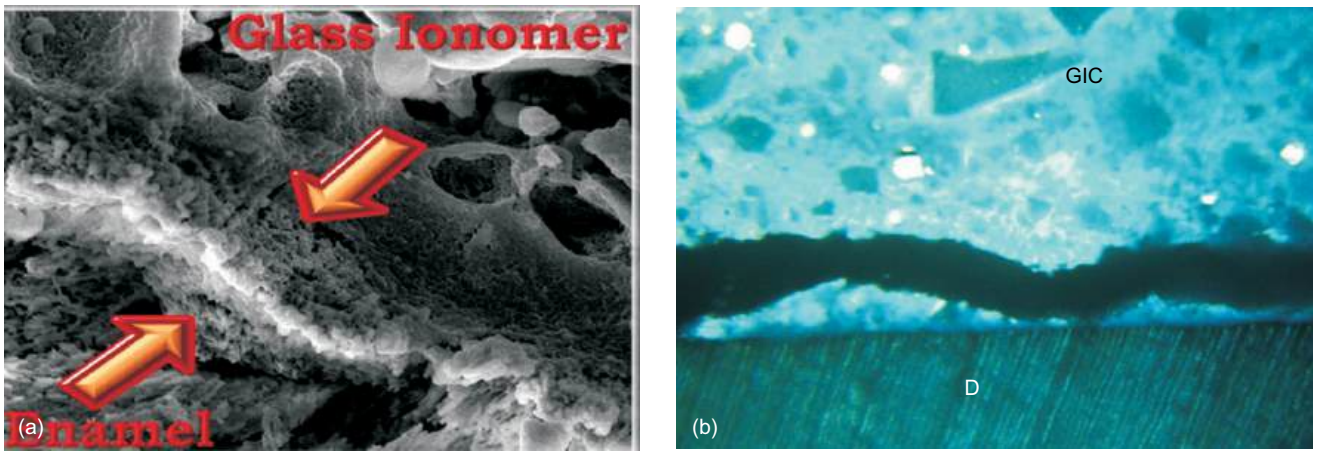


Figure 7.11 **(a)** Scanning electron microscopy image showing the interface between glass ionomer and enamel. The section surface has been etched lightly and a raised area (between the arrows) shows the *ion-enriched* layer at the tooth–restoration interface (field width 10 µm). **(b)** Fractured GIC–tooth interface (D) showing the cohesive failure common to these materials. The *ion-enriched* layer is still attached to the tooth (field width 300 µm). (Figure 7.11 (a) courtesy of Professor Hien Ngo.)

Q7.5: What is responsible for the cohesive fracture in Figure 7.11 (b)?

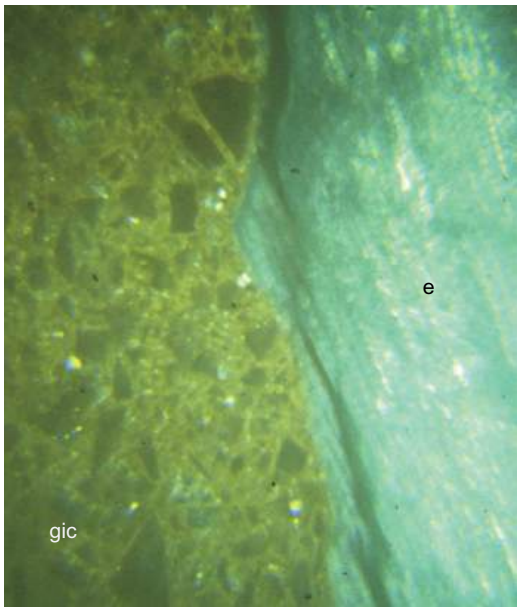


Figure 7.12 Photomicrograph of glass ionomer cement having pulled apart enamel prisms and caused cohesive failure in the enamel (e; black arrow) (field width 100 µm).

Q7.6: What are the irregular-shaped particles within the GIC in Figure 7.12?

prisms have been damaged/fractured/pulled apart in creating the cavity, the mineralized component of the interface can also fail cohesively due to the initial shrinkage stresses generated by the setting GIC (see Figure 7.12).

Dentine–collagen

Adhesion to collagen may occur through hydrogen bond formation or metallic ion bridging between carboxyl groups of the polyacid and collagen.

Dentine–tubules

There is limited evidence to argue for any significant micro-mechanical retention by the penetration of dentine tubules. The penetration is slight (< 5 µm), and the tensile strength of GIC in this proportion is poor, so is unlikely to contribute to retention.

Conditioning

Conditioning is the process by which both enamel and dentine are 'freshened' or pre-activated prior to the placement of freshly mixed GIC. The acid used is 10% polyacrylic acid for 10 seconds, which is then washed off and the surface dried gently, in order to:

- remove/modify the smear layer and expose calcium and phosphate ions on the mineralized surface
- increase the surface energy of the tooth surface to allow improved wettability of the GIC as it is placed on the tooth surface.

Dentine conditioner is mildly acidic, so it neither demineralizes the dentine excessively, nor opens up dentine tubules. Note that this conditioning step is not the same as the 'conditioners' used in resin composite retention—that material, even though sharing the same nomenclature (especially in the USA), is a much stronger acid etch (37% orthophosphoric acid).

7.3.4 Clinical uses of GIC relating to its properties

GICs are used conventionally as:

- luting cements for indirect restorations, where the powder:liquid ratio is closer to 1.5:1 (lower viscosity to improve cavity margin adaptation)
- direct restorative material, with a powder:liquid ratio of > 3:1 (stiffer, high-viscosity material)
- indirect pulp protection, with a powder:liquid ratio of 1.5:1, covering a direct pulp capping agent (e.g. calcium hydroxide or MTA) or in the depths of a cavity with very close proximity to the pulp, beneath an amalgam restoration (see Chapter 5, Section 5.11).

As a restorative material, the best physical properties (tensile and compressive strengths) of GIC are exhibited when the high-viscosity material is used in thick section, but GICs are still not universally recommended for *long-term* stress-bearing restorations (e.g. posterior occlusal restorations). Internal cavity line angles should be smooth and rounded prior to placement.

- *Fracture resistance*: original GICs were brittle and prone to fracture under heavy occlusal loading. Modern higher-viscosity derivatives have improved fracture toughness, but there is still little clinical evidence and justification for them being used routinely as *long-term* restorations in the posterior load-bearing dentition.
- *Abrasion resistance*: again generally poor, but modern resin-based surface coating systems (e.g. G-Coat Plus, GC Japan) can help to regulate the long-term water uptake and provide a protective and aesthetic coating, so improving this quality.
- *Physical properties*: compressive shear strengths, thermal expansion, and diffusivity have all improved with developments in GIC science. Modern materials appear to have some properties approaching more closely those of natural dentine.
- *Fluoride release*: fluoride ions are released initially at high levels, which fall away 8–10 weeks after placement. The fluoride ions are captured within the siliceous hydrogel matrix and can pass in and out of this to the tooth surface, so prompting the description that it can act as a 'fluoride reservoir', recharging from the high doses of professional fluoride application. There is contrary evidence, however, which argues that the fluoride release effect is of subclinical value after initial placement.
- *Aesthetics*: improvements have been made with better shade and translucency of modern GICs. Water uptake and high solubility affect the long-term stability of colour and translucency in restorations, but coating the surface just after placement or veneering the surface with composite at a later date are potential solutions.

7.3.5 Developments

As with dental resin composites, GIC material science is continuously evolving. Developments include the following:

- the speed, control, and conversion rate of the initial set, with fast-setting materials on the market taking 90 seconds to reach a carvable stage
- improving the physical properties in order to enable more reliable use in posterior, load-bearing cavities with the potential use of ceramic nano-filler technology or developing the use of N-vinyl pyrollidone-containing polyacids (NVPs) in the matrix polymer
- improving the wear resistance, polishability, and aesthetics of the final restoration with advances in resin-based coating systems, applied after finishing the restoration
- greater use of the ionic-exchange potential to introduce other ions to help to remineralize, repair tooth structure, and confer antibacterial effects.

7.4 Resin-modified glass ionomer cement (RM-GIC) and poly-acid modified resin composite ('compomer')

The description of these materials as 'light-cured GICs' should be avoided.

7.4.1 Chemistry

RM-GICs

RM-GICs essentially have the same chemistry as a conventional GIC with the addition of the hydrophilic resin, HEMA (hydroxyethyl-methacrylate), bis-GMA, and other photo-initiators. They set with a combination reaction: acid–base between the glass particles and polyalkenoic acid (as previously described) and also light-cured polymerization reaction of the resin (similar to that of a resin composite). The resin in RM-GICs can also undergo chemical polymerization due to an intrinsic redox (reduction–oxidation) reaction, so auto-curing over time (approximately 1 month) into a fully set material. Due to the fact that the HEMA resin is hydrophilic, there is a risk that after the initial snap-setting process, water is absorbed which might lead to medium- to long-term degradation and colour instability when used as the definitive restorative material (see Figure 7.13). These materials at a low powder:liquid ratio form the basis of widely used luting systems in fixed prosthodontics. Clinical examples include Fuji II LC®, Vitremer®, and Ketac Nano®.

Poly-acid modified composites

These are materials that are primarily resin-based, with some GIC chemistry incorporated, but not enough to promote an acid–base setting reaction. They essentially require light activation to promote the polymerization chain reaction. They cannot adhere chemically to tooth structure as do conventional GICs, as the acid–base reaction only occurs over a prolonged time period, after the initial set of the resin component. Therefore dentine bonding agents are a prerequisite for retention of these materials within a cavity. The original example of this class of material is Dyract®.

7.4.2 Clinical indications

Both of these materials can be used as provisional or definitive shade-matched tooth-coloured adhesive restoratives. However, there is some evidence that the colour stability of RM-GICs might be suspect over longer periods, due to the extent of water absorption that occurs (see Figure 7.13).

RM-GICs may also be used as a base in an adhesive *layered/laminate/'sandwich'* restoration with an overlying composite completely covering the RM-GIC base (a closed restoration). However, there is clinical evidence that these types of restoration are more prone to long-term failure, especially at the interfaces between the different materials. It is the authors' opinion that the *layered/laminate/'sandwich'* restoration should be limited to the placement of a resin composite veneer over the exposed occlusal surface of large GICs/RM-GICs at least 6 months after the GIC restoration has been placed to allow for its full maturation, so maximizing its physical and chemical properties.

When indirect chair-side or laboratory-made restorations are used, some form of cementation (or luting) is required (see Section 7.7.3), and the use of self-etching low-viscosity derivatives of the RMGIC technology has become popular for such an application. Although the adhesive bond strengths of these materials may be inferior to those of a multi-stage light- and self-cured resin adhesive system, used in conjunction with low-viscosity dual-cured composite luting cement their handling simplicity will produce more reliable everyday clinical performance. An example of this type of material is RelyX Unicem® (3MESPE).



Figure 7.13 **(a)** RM-GIC buccal cervical Class V restoration UL1 with good aesthetic properties initially. **(b)** The same restoration 7 years later, showing obvious intrinsic discoloration due to water absorption.

7.5 Dental amalgam

Dental amalgam (one of the oldest direct plastic restorative materials still in use) is an alloy of one or more metals including silver (Ag), tin (Sn), zinc (Zn), copper (Cu), and tiny amounts of some minor elements (palladium, platinum, indium) with mercury (Hg). Its use is rapidly on the decline worldwide as:

- Patients demand more aesthetic restorations and can appreciate the benefits of simple tooth-preserving restore/repair cycles of minimally invasive treatment.
- Health and safety and environmental concerns are raised about the use and disposal of mercury. This can be mitigated against by the use of appropriate traps in surgery waste water systems (compulsory in the UK). The United Nations Environment Programme (UNEP) 2013 Minamata Convention on Mercury has declared that the use of dental amalgam will be phased down, as caries prevention strategies and research into alternative restorative materials develop.
- Adhesive materials have developed with significantly improved bonding capabilities to sound and caries-affected tooth structure.
- Principles of minimally invasive operative dentistry have been embraced, producing smaller, less mechanically retentive cavities to restore, so promoting the use of adhesive materials.
- Caries control/management strategies have evolved with a better understanding of the histopathology of the disease.

7.5.1 Chemistry

- Silver (75–70%), tin (27–29%), copper (< 7% low Cu, > 12% high Cu), and trace metallic elements (< 1%) are presented as phases—γ gamma (Ag_3Sn), ε epsilon (Cu_3Sn), and d dispersant (Ag-Cu eutectic alloy)—and mixed with mercury (triturated in an amalgamator unit) to form an alloy (amalgamation reaction), so producing the intrinsic strength and mechanical properties of the set material.
- Metal phases are presented as ingots which are then made into lathe-cut particles. Alternatively, molten alloy is sprayed into an inert atmosphere and atomized into spherical particles. The final powder may be an admix of the two particle types. The particle size and shape have an effect on the ease of packing of the triturated amalgam into a cavity; spherical particle alloys require less force to condense into a cavity, resulting in fewer voids.
- Copper: modern alloys are high Cu alloys with > 12% Cu (single composition or dispersion-modified Cu-enriched alloys). The inclusion of copper improves corrosion and creep resistance and strengthens the set material by reducing unwanted Sn-Hg γ_2 (gamma 2) phase in the set amalgam.
- Zinc: > 0.01% Zn was introduced to scavenge oxygen when casting the metal ingots. The use of an inert atmosphere during the manufacture of spherical particles now makes the inclusion of zinc redundant. Inclusion of zinc led to increased hygroscopic expansion due to water contamination during placement, but may increase marginal strength.

7.5.2 Physical properties

- *Strength*: high compressive and tensile strengths several days after placement. Care is required when checking the occlusion immediately after condensation and carving, as the amalgam may be brittle at this stage. High Cu alloys have a greater strength and are used in large load-bearing posterior restorations. Amalgam is weak in thin section and requires at least 2 mm thickness to support itself. Therefore cavity design is important for this reason as well as for mechanical retention via undercuts and for support.
- *Corrosion*: an electrochemical breakdown due to the interaction between any metal and its surroundings. Low Cu amalgams corroded quickly and extensively, so sealing the micro-gaps between the cavity wall and restoration. High Cu alloys are more resistant to corrosion and corrosion fatigue at the margins. Amalgam surfaces can oxidize over time, leading to surface corrosion or even breakdown, resulting in pitting and surface deficiencies.
- *Rigidity:* the modulus of elasticity of amalgam is high but not as rigid as enamel.
- *Dimensional changes:* after minor setting contraction, water uptake during setting (especially in low Cu, Zn-containing alloys) can lead to hydrogen production and expansion, resulting in pain. Thermal expansion can also occur to a degree further than normal tooth structure, so fine surface polishing, the friction of which will generate heat, should be undertaken with due care.

7.5.3 Bonded and sealed amalgams

Amalgam restorations obtain retention within a cavity macro-mechanically, achieved by making undercut preparations (where the base of the cavity is wider than its opening) or using slots/grooves, root canal posts, or the pulp chamber and root canal orifices themselves (Nayyar core; see Figure 7.14). All of these require excessive amounts of tooth structure to be removed or damaged to some extent, thus ultimately weakening the remaining tissue—not particularly minimally invasive! Dentine pins have been used to retain amalgam in larger cavities where cusps are missing. There is clear evidence that correct placement of dentine pins tends to significantly increase the stresses in an already weakened tooth, leading to ultimate catastrophic failure of the restoration and tooth. It is the authors' opinion that dentine pins should not be used in modern dentistry, as alternative methods and materials now exist, including self-curing adhesive luting cements to couple the cavity walls to the amalgam restoration.

Bonded amalgams

An adhesive, chemically cured, resin-based luting cement is used to seal the exposed dentine tubules after cavity preparation, bonding to the dentine mineral and collagen (as with resin composites) and micromechanically interlocking simultaneously with the freshly condensed amalgam as they both set (see Chapter 8, Section 8.9). Chemical bonds with metal oxides forming in the set amalgam may form, but these have



Figure 7.14 Left bitewing radiograph showing an amalgam Nayyar core LL7. Note how the 'amalgam pins' extend into the coronal aspects of the endodontically treated root canals, to aid retention after the gutta-percha was removed. The bulk of amalgam will also confer strength on the restoration. For the clinical steps to place a Nayyar core, see Chapter 8.

Q7.7: What problem can you see with the restoration in Figure 7.14 and how can it be treated? What do you think should be considered as part of the long-term management of this tooth?



Figure 7.15 Broken down mandibular molar restored with a large load-bearing amalgam restoration. (Courtesy L. Mackenzie.)

Q7.8: At what stage of restoration placement has the photograph in Figure 7.15 been taken, and what stage is left to complete?

little clinical significance for retention in the long term. The evidence for their routine use is inconclusive, but the procedure might be beneficial when repairing old fractured large amalgams with new amalgam.

Sealed amalgams

As modern high copper alloys corrode less and so do not adequately seal the spaces at the tooth–amalgam interface, a flowable resin/dentine bonding agent can be used to coat the surface of the tooth–amalgam interface after condensation, carving, and finishing. This can penetrate the micro-gaps at the marginal interface which form during the setting process, so reducing the likelihood of marginal leakage and the risk of subsequent caries. A dentine bonding agent may also be used to seal the dentine cavity walls prior to placement of the amalgam restoration.

7.5.4 Modern indications for the use of amalgam

In the current climate of more acceptable tooth-coloured alternatives and minimally invasive dentistry, as well as the ever growing environmental concerns about the safety and disposal of mercury, the use of dental amalgam is becoming more limited. Due to the presence of alternatives, the uses of amalgam may reasonably be restricted to large posterior load-bearing restorations or as a core material for crowns (see Figure 7.15 and Chapter 8).

7.6 Temporary (intermediate) and provisional restorative materials

7.6.1 Characteristics

Temporary restorative materials should ideally be:

- simple for both dentist and nurse to handle clinically
- easy to remove to allow final placement of the definitive restoration
- materials that do not interfere with the setting/bonding chemistry of any definitive restorative material
- durable enough to last in the oral cavity for several weeks
- cheap and biocompatible.

Provisional restorative materials should have all of the characteristics described in the previous list, and ideally they should be:

- durable enough to last in the oral cavity for several months
- as aesthetic as possible.

The boundary between a provisional and definitive restorative material is nowadays quite blurred, as many adhesive restorative materials combine both functions.

7.6.2 Chemistry

- Radio-opaque, polymer-reinforced zinc oxide:eugenol cements (e.g. Kalzinol®, Sedanol®, pre-mixed Intermediate Restorative Material, or IRM®). Quick setting and easy to spatulate and load into cavities, but must not be used where resin composites will follow, as the eugenol affects the polymerization chain reaction adversely.
- Glass ionomer cements (see previous sections).
- Zinc polycarboxylate cements (Poly F® Plus): water-soluble, low-molecular-weight polymers of acrylic or methacrylic acid that form solid, insoluble products when mixed with specially prepared zinc oxide powder. The resulting cement adheres to dental enamel and can also be used as a luting agent.

7.7 Calcium silicate-based cements

7.7.1 History

Calcium silicate-based cements were first introduced to operative dentistry in 1993, when Torabinejad developed a formula based on ordinary Portland cement (OPC), which was intended primarily for use as an endodontic material (e.g. in root end/perforation repairs). A mineral trioxide aggregate (MTA), which is composed principally of tri-calcium silicate, di-calcium silicate, tri-calcium aluminate, and tetra-calcium aluminoferrite, its operative dental applications were somewhat limited due to the protracted working and setting time of several hours.

In 2010, Biodentine™ (Septodont, France), a quicker setting calcium silicate-based restorative cement, was introduced to the market. This cement was developed as a dentine replacement material, a novel clinical application for this family of materials, with the intention that it would function as a direct coronal restoration. The comparatively short working/setting time (approximately 12 minutes) enables the use of this cement for clinical restorative procedures, which would be impossible with modern MTAs that achieve only an initial set within 3–4 hours.

7.7.2 Chemistry and interactions with the tooth

The faster-setting calcium silicate cement (Biodentine™) is composed principally of a highly purified tri-calcium silicate powder that is prepared synthetically in the laboratory *de novo*, rather than being derived from a clinker product of cement manufacture. In addition, Biodentine™ contains di-calcium silicate, calcium carbonate, and zirconium dioxide as a radio-opacifier. The powder is dispensed in a two-part capsule to which is added an aliquot of hydration liquid, composed of water, calcium chloride (to accelerate the setting reaction), and a reducing agent.

Similar to OPC, calcium silicate-based dental cements set via a hydration reaction. Although the chemical reactions taking place during the hydration are more complex, the conversion of the anhydrous phases into corresponding hydrates can be simplified as follows:

$$2Ca_3SiO_5 + 7H2O \longrightarrow 3CaO.2SiO_2.4H_2O + 3Ca(OH)_2 + \text{energy}$$

$$\textbf{C3S} + \textbf{water} \longrightarrow \textbf{CSH} + \textbf{CH}$$

$$2Ca_2SiO_4 + 5H_2O \longrightarrow 3CaO.2SiO_2.4H_2O + Ca(OH)_2 + \text{energy}$$

$$\textbf{C2S} + \text{water} \longrightarrow \textbf{CSH} + \textbf{CH}$$

This setting reaction is a dissolution–precipitation process that involves a gradual dissolution of the unhydrated calcium silicate phases (C_3S and C_2S) and the formation of hydration products, mainly calcium silicate hydrate (CSH) and calcium hydroxide (CH). The CSH precipitates as a colloidal material and grows on the surface of unhydrated calcium silicate granules, forming a matrix that binds the other components together, gradually replacing the original granules. Meanwhile, calcium hydroxide is distributed throughout the water-filled spaces present between the hydrating cement species.

The setting cement attains a high pH and this has a novel effect on the dentine. Any organic material that is exposed (e.g. demineralized collagen in caries-affected dentine) will be 'caustically etched' by the cement and then subjected to calcium ion release from the cement. This alkaline environment is very conducive to mineralization within the interface, so this material can truly be termed 'bioactive.' Equally important is the therapeutic effect on the pulp elicited by the release of calcium hydroxide, leading to reparative (tertiary) dentine formation.

7.7.3 Clinical applications

The presence of large amounts of calcium hydroxide within the set material make this material potentially useful for its therapeutic pulp effects, being more robust and resistant to dissolution compared with the traditional calcium hydroxide pulp protection materials. The fast-setting calcium silicate cement described here has sufficient compressive strength to act as a bulk replacement for dentine, but is not appropriate for long-term exposure to wear or occlusal loading in the mouth. As such, it has an appropriate use as an intermediate/provisional restorative material, especially when restoring deep carious lesions (see Figure 7.16 and Section 8.7), prior to reducing the cement height and veneering the occlusal and proximal surfaces with a material such as resin composite. The relatively slow set and maturation of this material preclude the placement of the resin composite veneer immediately on to the newly placed, initially set cement. It is better to leave it for at least 1 week to mature.

- The cut surface of the hard cement should be given a 15-second acid etch, either with phosphoric acid (followed by thorough water rinsing) or using a self-etching adhesive system, treating the cement surface much as one would dentine.
- The interface between the overlying resin composite and cement will be micro-mechanical in nature, in a similar manner to the interface between GICs and resin composites.
- The potential for the long-term acid dissolution of exposed cement under a biofilm on a proximal restoration surface would indicate that the 'closed-sandwich' type of restoration would be favoured where this is clinically achievable (see Figure 7.16).

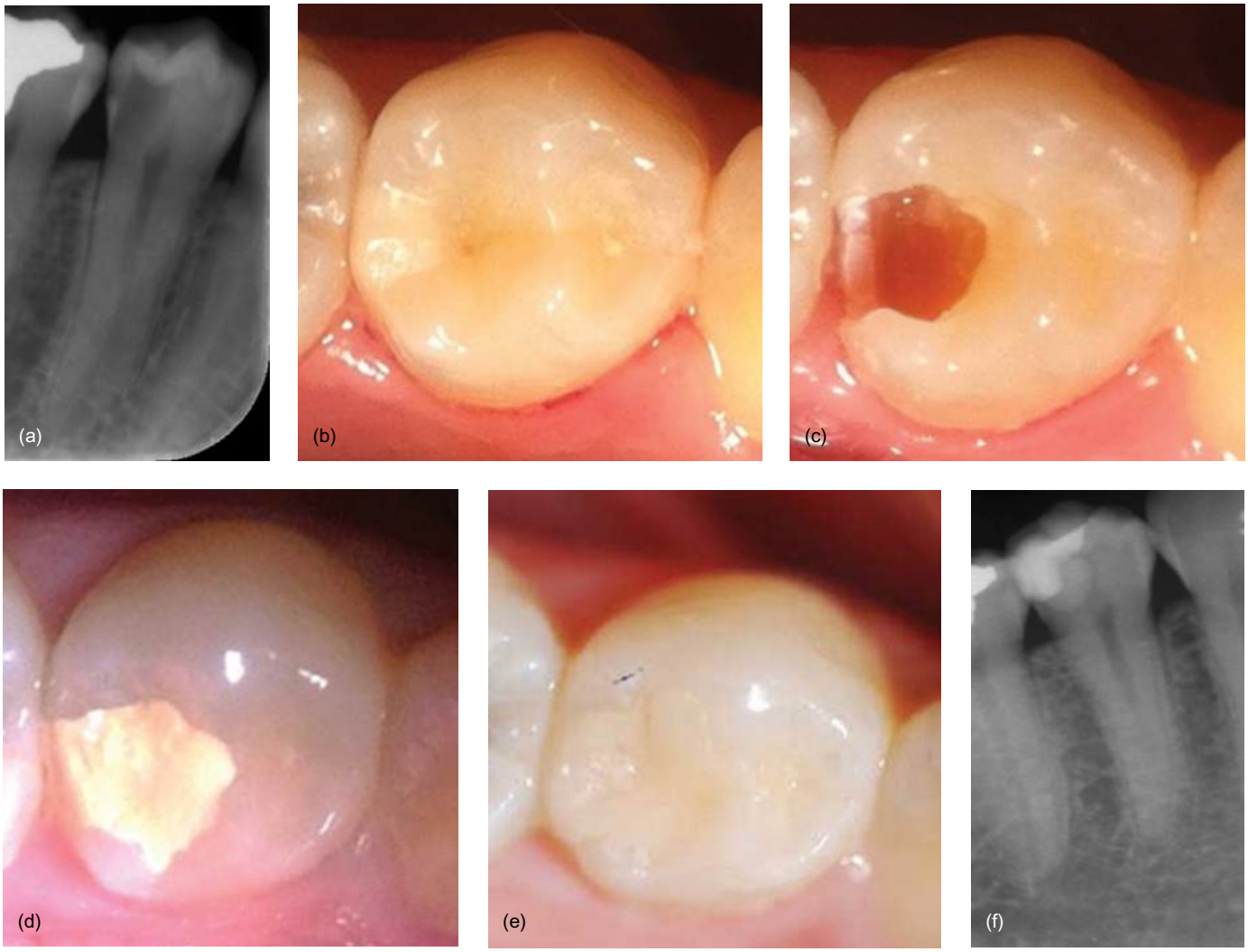


Figure 7.16 **(a)** Long cone periapical radiograph of an LR4 with a deep carious lesion distally. **(b)** The tooth was vital to the electric pulp test and exhibited the classic symptoms of an acute reversible pulpitis. A clinical decision was made to restore it minimally invasively as it was cavitated distally. **(c)** The same tooth with the lesion accessed from the occlusal aspect. Dental burs and then Carisolv™ gel were used to excavate the deepest carious dentine. No pulp exposures were created. No cavitation was observed on the mesial surface of the LR5, so this was treated non-operatively with fluoride varnish application as a preventive measure. **(d)** The Biodentine™ restoration after 1 month *in situ*. It can be seen how normal functional occlusal wear and tear has taken its toll on the restoration surface, as it is roughened significantly. **(e)** The same tooth after receiving a resin composite veneer restoration overlying the Biodentine™ beneath. This is 1 year later. **(f)** The review periapical radiograph at that time, showing no PA changes or further progression of disease. The LR4 has maintained its vitality, and the original presenting symptoms subsided fully within 1 week of the Biodentine™ being placed. Note, however, the thin radiolucent line beneath the Biodentine™ restoration in the radiograph.

Q7.8:

i What other management regimes would you put in place to manage the patient in Figure 7.16?

ii Look at Figure 7.16 (a). What signs are evident on the radiograph that the LR4 has been resisting the caries process as it has gradually progressed through the tooth?

iii Look at Figure 7.16 (b). What are the classic symptoms of an acute reversible pulpitis? Do you think this distal lesion would be easy to detect on clinical examination?

iv Look at Figure 7.16 (c). What is Carisolv™ gel? How does it work? What quality of enamel and dentine has been retained at the base of this cavity and why?

v Look at Figure 7.16 (f). Would you describe the resin composite veneer overlying the Biodentine™ as 'open sandwich' or 'closed sandwich'? What do you think might be the cause of the radiolucent line in the radiograph beneath the Biodentine™ restoration?

7.8 Materials and techniques for restoring the endodontically treated tooth

7.8.1 Materials

Root canal posts can be made using metals such as stainless steel, titanium, and gold, and can be directly fabricated chair-side by the dentist or indirectly cast in the laboratory. These techniques and materials have been used for many years. More recently, fibre-reinforced posts have also become popular. These posts consist of very fine glass fibres extruded in the long axis of the post and shaped within a matrix of resin, normally an epoxy or methacrylate which is then fully polymerized, but retaining a roughened surface to aid retention of the cement luting material. The shape of the post is designed to mimic the tapered anatomy of the enlarged root canal space, thus reducing the need for unnecessary root dentine removal during placement. Clinical studies have reported that direct fibre-reinforced composite posts out-perform cast and metal posts in general. However, the evidence cannot be considered conclusive.

Compared with metal posts, fibre posts (due to their flexibility and closer match to the elastic modulus of dentine) would seem to protect the root against fracture, especially when there is extensive coronal tissue breakdown/loss. The most common type of failure with fibre-reinforced composite posts is de-bonding, which normally permits re-restoration of the tooth. The current available evidence does not rule out the use of cast posts, fabricated in the dental laboratory. However, since the use of cast posts may result in a significantly greater loss of tooth structure during preparation and on failure, compared with fibre posts, their application should be limited to those cases in which no additional dentine has to be removed to allow for their cementation.

7.8.2 Root canal post cementation

Fibre and metal posts can be cemented into the root canal using all types of adhesive systems. The use of conventional two- and three-step bonding systems can be challenging, due to the difficulty in light curing the adhesive within the depths of the root canal space. Polymerization shrinkage stresses are also likely to develop in the root canal, which is a very confined space with an unfavourably high C factor (see Chapter 6, Section 6.3.1). For this reason, self-etching adhesive cements based on resin-modified glass ionomer cement technology have become very popular, as they are easy to use, create a uniform layer of cement along the root canal walls, and still allow the use of conventional two- and three-step adhesive systems to more effectively bond the resin composite core to the residual coronal tooth structure.

7.9 Suggested further reading

Darvell BW (2009) *Material Science for Dentistry*, 9 th edn. Cambridge: Woodhead Publishing Ltd. <**www.woodheadpublishing.com/en/book.aspx?bookID=1547**>

Curtis RV, Watson TF (eds) (2008) *Dental Biomaterials: imaging, testing and modelling*. Cambridge: Woodhead Publishing Ltd. <**www.woodheadpublishing.com/en/book.aspx?bookID=1347**>

7.10 Answers to self-test questions

Q7.1: What properties of the resin composite are affected by monomer chain length?

A: The viscosity of the uncured material and the degree of polymerization shrinkage (see text).

Q7.2: A dentine smear layer is illustrated in Figure 7.6. How is the smear layer created?

A: By the friction generated from any cutting instrument on the tooth surface (hand instrument or rotary). Air abrasion produces a layer similar to a smear layer created from organic/inorganic chip debris and abrasive particles.

Q7.3: What are the dark irregular shapes outlined within the rhodamine-labelled (red) upper portion in Figure 7.8 (c)?

A: These are the irregular silica-based filler particles within the resin composite.

Q7.4: Look at Figure 7.10 (a) and (b). Can you spot the difference between the two and explain the reason for the difference?

A: Note the dark discoloration on the superior margin of the restoration–enamel interface. This is due to the enamel not being as efficiently etched by the weaker self-etching primer. This has led to a less well-sealed adhesive bond, which has gradually picked up stain over the last decade of use. Pre-etching the enamel with 37% orthophosphoric acid etch can alleviate this problem.

Q7.5: What is responsible for the cohesive fracture in Figure 7.11(b)?

A: Phase 2 of the setting reaction with the formation of calcium and aluminium cross-links coupled with potential dehydration of the GIC.

Q7.6: What are the irregular-shaped particles within the GIC in Figure 7.12?

A: Unreacted fluoro-calcium aluminosilicate glass particles.

Q7.7: What problem can you see with the restoration in Figure 7.14 and how can it be treated? What do you think should be considered as part of the long-term management of this tooth?

A: Note the distal amalgam overhang, caused by inadequate adaptation of the matrix band when originally packing the cavity. This may be removed using fine burs (if clinical access is available) and interproximal amalgam finishing strips. If not, the long-term management of the tooth may involve the placement of an indirect full coverage cast restoration (e.g. a metal ceramic or full gold crown), and during the tooth preparation for this the overhang will be automatically removed.

Q7.8: At what stage of restoration placement has the photograph in Figure 7.15 been taken, and what stage is left to complete?

A: Just after the amalgam has been packed, circumferential matrix band removed, and the restoration finished. This means that the margins, contour, surface finish, contact points, and occlusion have all been checked. The next stage might be to polish the restoration to smooth the surface, thereby reducing potential plaque accumulation. However, polishing techniques must use water cooling to avoid an increase in frictional heat causing damage to the pulp (see Figure 6.1(b) in Chapter 6 for the final result).

Q7.9:

i What other management regimes would you put in place to manage the patient in Figure 7.16?

A: A preventive regime consisting of OHI, dietary advice, and use of fluoride. Ensuring that the caries risk of this patient is brought under control, with regular review consultations to assess the patient's adherence to the behavioural change advice given by the dental team, as well as to check the progress (or not) of further disease.

ii Look at Figure 7.16 (a). What signs are evident on the radiograph that the LR4 has been resisting the caries process as it has gradually progressed through the tooth?

A: Note the reduced size of the pulp chamber directly adjacent to the lesion. This is tertiary dentine that has been laid down in response to the gradual caries process.

iii Look at Figure 7.16 (b). What are the classic symptoms of an acute reversible pulpitis? Do you think this distal lesion would be easy to detect on clinical examination?

A: Short, sharp pain of a few seconds' duration, stimulated by hot/cold/sweet stimuli, and poorly localized. Clinically, this lesion may be difficult to detect on visual examination. However, even the wet surface shows a slight opacity on the distal occlusal aspect—drying the tooth would make this more evident. Careful probing proximally may also help to detect the surface roughness of the cavitated lesion. This shows the importance of bitewing radiographs in the examination of such at-risk patients.

iv Look at Figure 7.16 (c). What is Carisolv™ gel? How does it work? What quality of enamel and dentine has been retained at the base of this cavity and why?

A: Read Section 5.8.5 in Chapter 5 for revision about chemo-mechanical excavation gels. There is caries-affected dentine retained on the cavity base. The enamel margin is sound (it appears slightly frosty as it has been scratched by a gingival margin trimmer hand instrument to help to gauge its strength). If the cavity was to be excavated to sound dentine (yellow and hard) there would be a significant risk of unnecessary pulp exposure, and the restoration margin would end up being very subgingival, thus calling into question the actual restorability of the tooth/the viability of placing a restoration.

v Look at Figure 7.16 (f). Would you describe the resin composite veneer overlying the Biodentine™ as 'open sandwich' or 'closed sandwich'? What do you think might be the cause of the radiolucent line in the radiograph beneath the Biodentine™ restoration?

A: This is a closed-sandwich restoration, as no Biodentine™ has been left exposed to the oral cavity. Please read Section 2.5.3 in Chapter 2 for the answer to the second part of this question. As Biodentine™ caustically etches the dentine, due to its high pH, it is possible that this radiolucency has been created by this process, and only time will tell (with review radiographs) if this gains mineral and therefore becomes more radio-dense like the surrounding sound dentine.